Project Report

On

"Social Media Usage & Screen Time Analysis"

Submitted in Partial Fulfillment for Degree of

"Bachelor of Computer Application"

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Date: _____

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1. Introduction:

The advent of smartphones has integrated social media and digital screens deeply into daily life. While beneficial for connectivity, excessive screen time is increasingly linked to poor mental health, lack of sleep, and decreased productivity. This project analyzes a dataset containing user demographics and digital habits to uncover patterns between screen time, stress levels, and work productivity, providing data-driven insights into our modern digital behaviors.

2. Database Description:

The dataset used for this analysis (`socialmedia_data.csv`) comprises comprehensive user profiles, capturing critical metrics:

- **Demographics:** Age, Gender, Occupation.
- **Digital Habits:** Daily Phone Hours, Social Media Hours, Weekend Screen Time Hours, Primary Device Type.
- **Lifestyle & Wellbeing:** Sleep Hours, Work Productivity Score, Stress Level (1-10 scale).

3. Data Understanding:

Initial exploration of the dataset reveals wide variations in digital consumption across different demographic sectors. By analyzing average daily phone hours, social media hours, and sleep patterns, we establish a baseline understanding of "normal" usage and recognize the immediate trade-offs users make between screen time and essential recovery (sleep).

4. Exploratory Data Analysis (EDA):

EDA forms the foundation of this report, summarizing the main characteristics of the dataset. Key summary statistics provide a high-level view of the user base, laying the groundwork for more complex correlational and predictive modeling.

5. Handling Missing values & Outliers:

Before in-depth analysis, the dataset underwent strict preprocessing. The `clean_data()` function ensures data integrity by identifying and summarizing any missing values (nulls) and stripping the dataset of any duplicate user entries (`User_ID`), ensuring every data point represents a unique user profile.

6. Spread & Distribution of Data:

The dataset features a wide distribution across age groups, professional roles, and tech habits. To quantify this spread, Feature Engineering was applied to standardize certain metrics for analysis:

- **High Stress Flagging:** Transforming numerical stress scores into a binary "High Stress" label (Yes/No) for users scoring a 7 or above.
- **Estimated Weekly Screen Time:** A derived metric aggregating weekday and weekend hours to reflect total weekly digital exposure.

7. Automating EDA:

To streamline the analysis process, an interactive Python script (`**socialmedia_analysis.py**`) was developed. This script automates routine EDA tasks, allowing the user to select specific analyses (e.g., Usage by Occupation, Sleep vs. Stress Level) from a command-line interface, instantly computing and returning the relevant metrics without manual data manipulation.

8. Regression Analysis:

To predict continuous numerical outcomes, a Multiple Linear Regression model was implemented.

- **Goal:** Predict `Daily_Phone_Hours`.
- **Features Used:** Age, Gender, Occupation, Social Media Hours, Sleep Hours, and Stress Level.
- **Process:** Categorical variables were label-encoded. The dataset was split 80/20 for training and testing. The model identifies linear relationships, utilizing coefficients to determine the weight each feature has on predicting total daily phone usage.\

9. Supervised Learning Models:

Moving beyond descriptive statistics, the project leverages supervised machine learning techniques (algorithms that map an input to an output based on example input-output pairs). The two primary algorithms utilized in this project are Linear Regression (for continuous value prediction) and Random Forest Classification (for categorical classification).

10. Classification Task:

While regression predicted hours, the classification task aimed to predict an individual's state of mind.

- **Goal:** Predict if a user falls into the "High Stress" category (Yes/No).
- **Model:** Random Forest Classifier (an ensemble learning method utilizing multiple decision trees).
- **Features Used:** Age, Gender, Occupation, Daily Phone Hours, Social Media Hours, Sleep Hours.

11. Model Evaluation:

Both predictive models were rigorously evaluated against the 20% validation test set:

- **Regression Model Evaluation:** Evaluated using Mean Squared Error (MSE), which measures the average squared difference between estimated and actual values, and R-squared (R^2), which indicates the proportion of variance in phone hours explained by the model's features.
- **Classification Model Evaluation:** Evaluated using overall Accuracy (percentage of correct stress predictions), and a detailed Classification Report citing precision, recall, and f1-score for both stress classes.

12. Data Visualization:

Visual representations are crucial for digesting the processed data. The analysis script auto-generates five key charts:

1. **Average Daily Phone Hours by Occupation (Pie Chart):** Highlights which professions demand or encourage the most screen time.
2. **Device Type Distribution (Bar Chart):** Shows the prevalence of smartphones vs. laptops and tablets in everyday usage.
3. **Trend of Daily Phone Hours by Age (Line Chart):** Illustrates how screen time habits shift across different generational age brackets.
4. **Correlation of Numerical Features (Heatmap):** A color-coded matrix revealing strong statistical correlations (e.g., mapping inverse relationships like high screen time vs. low sleep).
5. **High Stress Prediction (Confusion Matrix):** A scorecard evaluating the Random Forest model, charting True Positives, True Negatives, False Positives, and False Negatives.

13. Results & Findings:

The comprehensive analysis yielded several significant findings:

- Individuals in specific occupations (often tech or corporate-aligned) exhibit substantially higher daily screen times.
- There is a distinct, observable inverse correlation between high screen time/social media usage and average sleep hours.
- Elevated stress levels are prevalent among users who sacrifice sleep for extended phone usage.
- The Random Forest model proved capable of establishing a baseline prediction for high-stress individuals based solely on their demographic data and digital habits.

14. Conclusion:

This project successfully demonstrates the profound impact digital consumption has on modern well-being. By automating the EDA process and applying predictive modeling, the analysis transitions from merely observing data to actively identifying behavioral risk factors. The findings suggest that monitoring and managing daily screen time can be a viable strategy for improving sleep quality and reducing overall stress.

15. Future Scope:

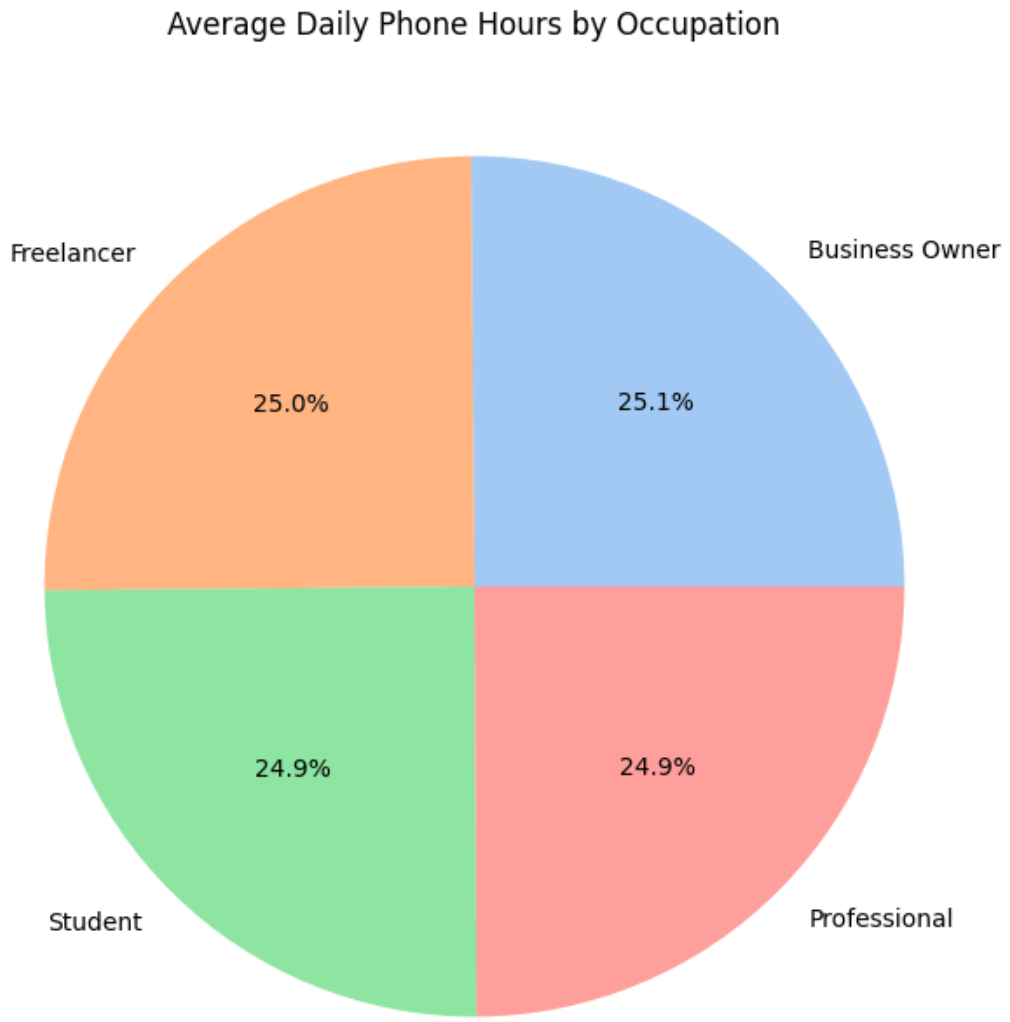
While current models provide valuable baseline predictions, future iterations of this project could be enhanced by:

- Integrating real-time API data from social media platforms or screen-time tracking apps.
- Applying more advanced Deep Learning models (e.g., Neural Networks) for higher accuracy stress prediction.
- Expanding the dataset geographically and culturally to account for divergent global digital habits.
- Creating a dynamic web dashboard (e.g., utilizing Streamlit or Dash) to replace the command-line interface, offering interactive visualizations for non-technical users.

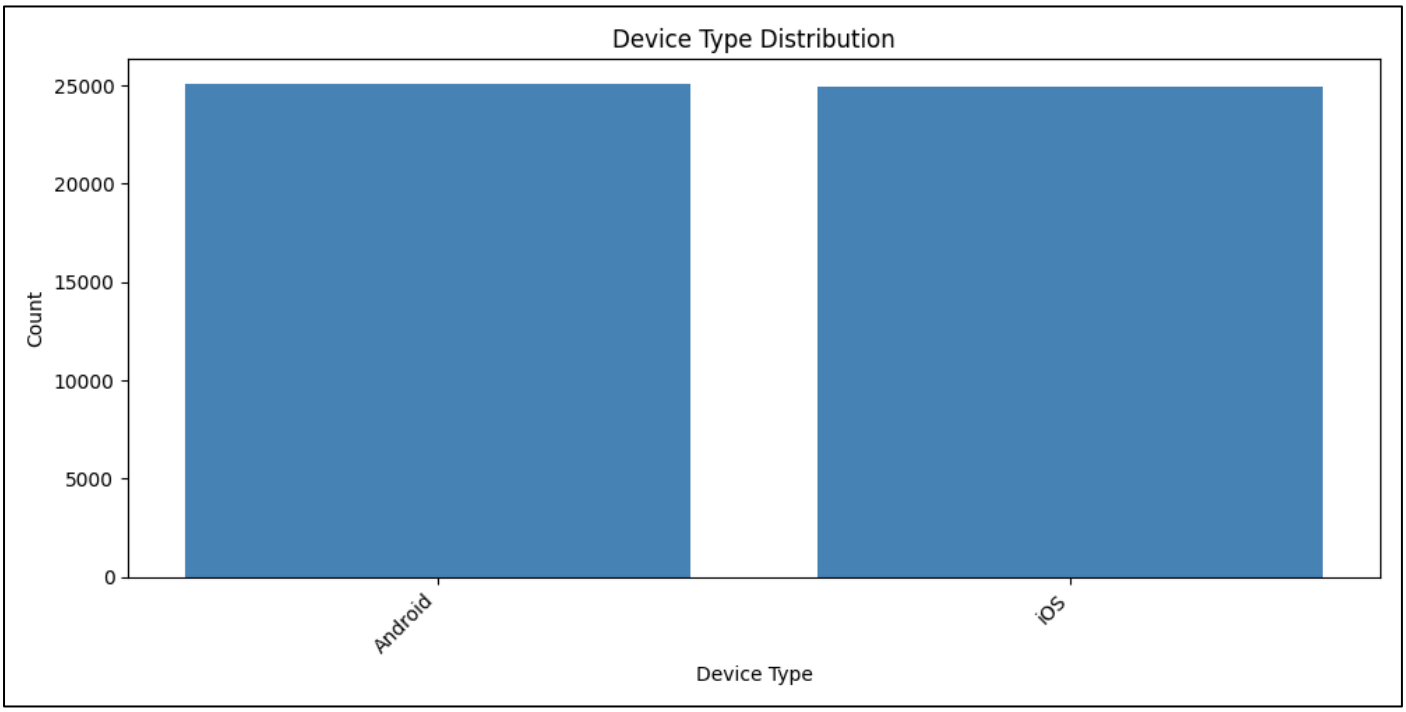
16. References:

Description	Link
Pandas (Data Manipulation	https://pandas.pydata.org/docs/
NumPy (Numerical Computing)	https://numpy.org/doc/
Matplotlib (Data Visualization)	https://matplotlib.org/stable/contents.html
Seaborn (Statistical Data Visualization)	https://seaborn.pydata.org/
Scikit-Learn (Machine Learning)	https://scikit-learn.org/stable/

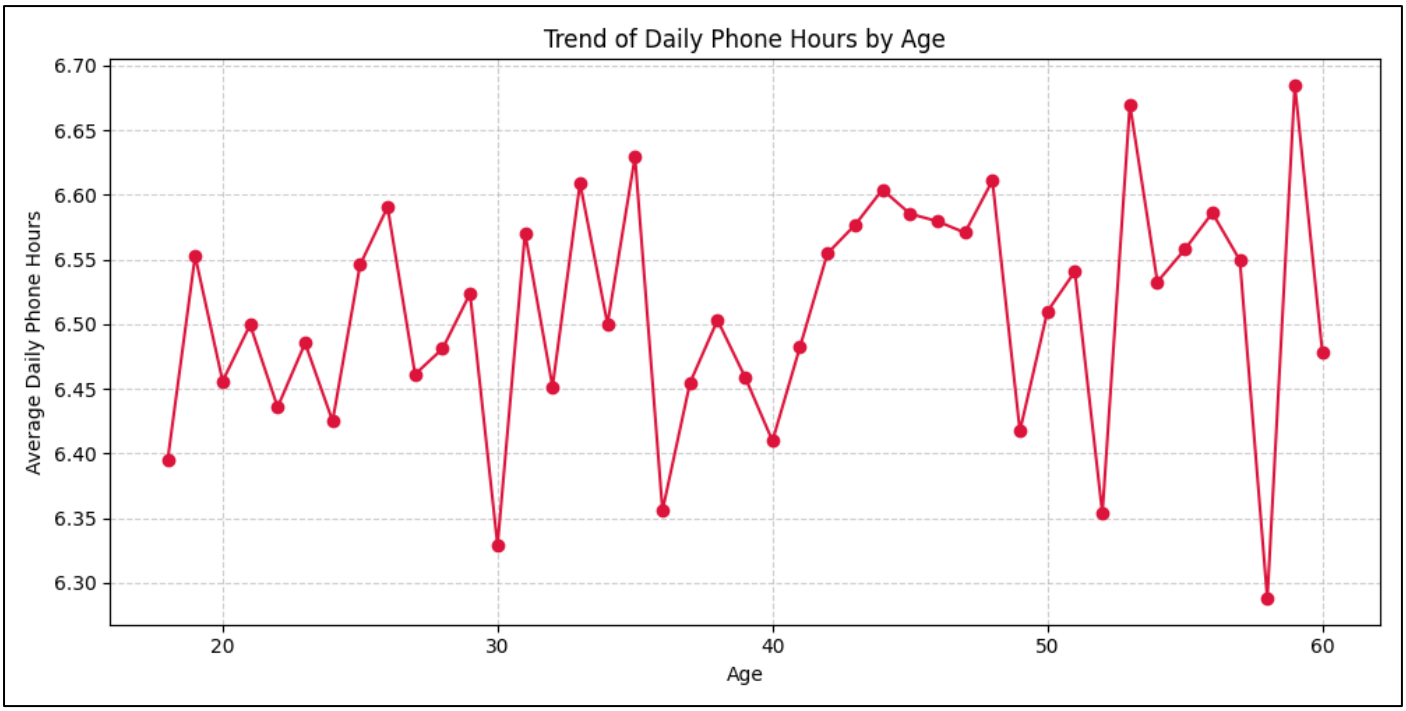
1). Average Daily Phone Hours by Occupation:



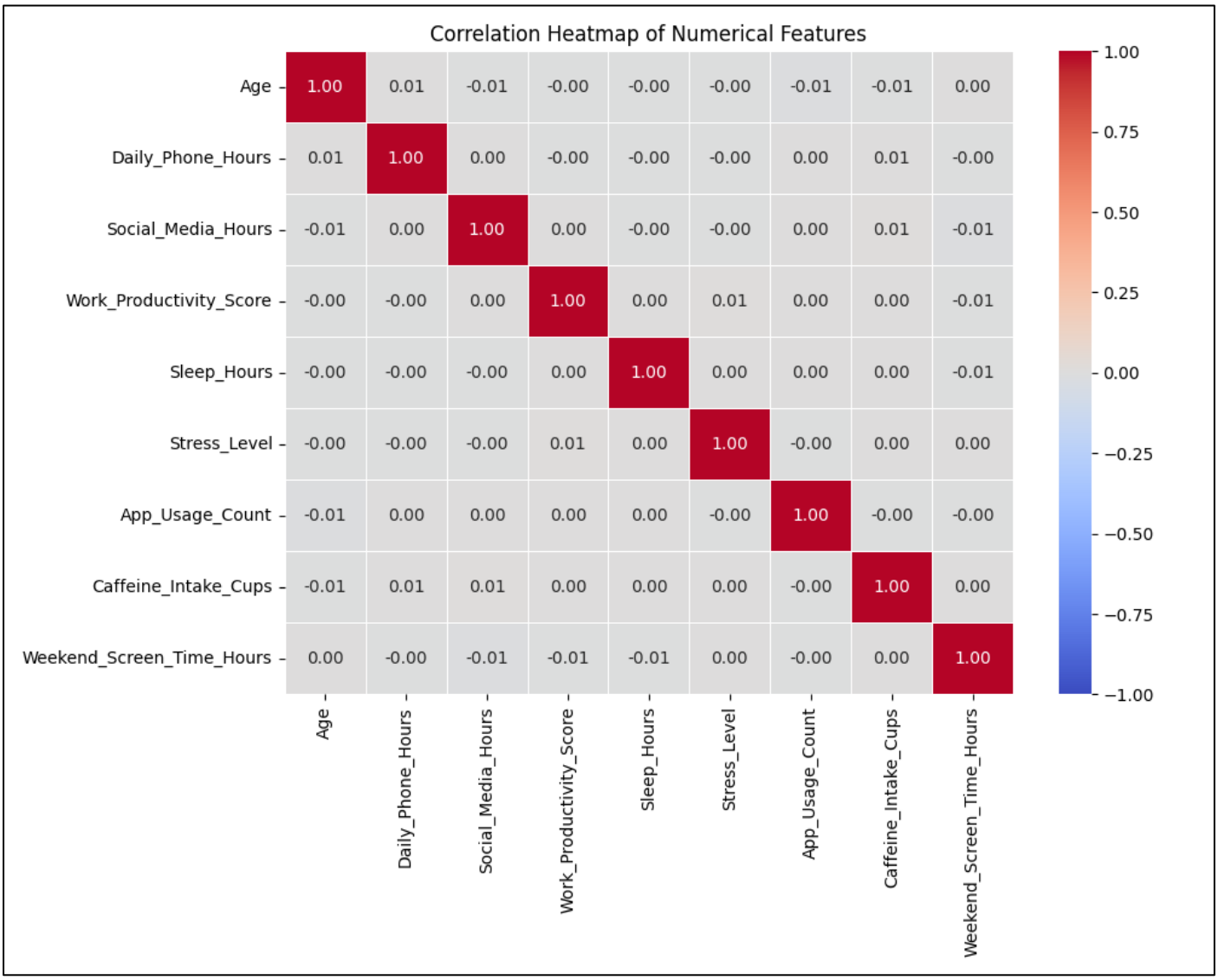
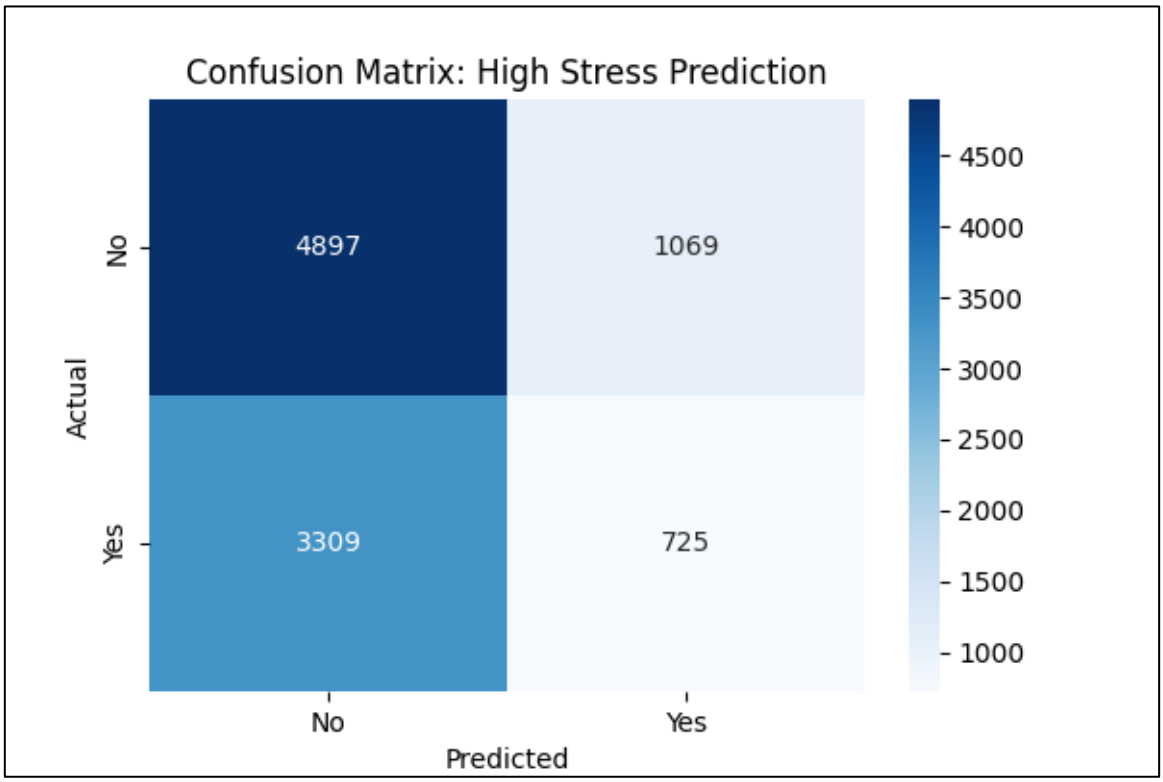
2). Device Type Distribution:



3). Trend of Daily Phone Hours by Age:



4). Correlation of Numerical Features (Heatmap):



5). High Stress Prediction: